Session 6 and 7 Assignment: Correlation, Regression, and ANOVA

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2026-09-28

setting a working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(dplyr)
library(rio)
library(tidyverse)
library(ggthemes)
library(ggplot2)
library(summarytools)
library(sjmisc)
library(readxl)
library(car)
library(lmtest)
library(nortest)
library(knitr)
library(kableExtra)
library(tidyr)
library(broom)
library(formatR)
```

Data Loading and Preparation

```
raw_data <- read_xlsx("ass4.xlsx", sheet = "Sheet1")
head(raw_data)
```

```
# A tibble: 6 × 7
     ID Department Work_Mode Years_Experience Training_Hours Productivity_Score
  <dbl> <chr>      <chr>                <dbl>          <dbl>              <dbl>
1     1 Sales      Onsite                   3             10                 58
2     2 Sales      Onsite                   5             15                 68
3     3 Sales      Onsite                   2              8                 52
4     4 Sales      Onsite                   7             20                 75
5     5 Sales      Onsite                   4             12                 62
6     6 Sales      Remote                   6             18                 78
# i 1 more variable: Stress_Index <dbl>
```

```
raw_data$ID <- as.character(raw_data$ID)
clean_data <- raw_data[grepl("^[0-9]+$", raw_data$ID), ]
clean_data$ID <- as.numeric(clean_data$ID)
clean_data <- clean_data[!duplicated(clean_data[, -1]), ]
cat("Number of rows after cleaning:", nrow(clean_data), "\n")
```

```
Number of rows after cleaning: 30
```

```
cat("Number of columns:", ncol(clean_data), "\n")
```

```
Number of columns: 7
```

```
kable(head(clean_data, 10), caption = "First 10 rows of cleaned data")
```

First 10 rows of cleaned data

ID	Department	Work_Mode	Years_Experience	Training_Hours	Productivity_Score	Stress_Index
1	Sales	Onsite	3	10	58	7
2	Sales	Onsite	5	15	68	6

3	Sales	Onsite	2	8	52	8
4	Sales	Onsite	7	20	75	5
5	Sales	Onsite	4	12	62	7
6	Sales	Remote	6	18	78	4
7	Sales	Remote	8	22	85	3
8	Sales	Remote	1	5	45	9
9	Sales	Remote	9	25	88	2
10	Sales	Remote	10	30	92	2

```
clean_data$Department <- as.factor(clean_data$Department)
clean_data$Work_Mode <- as.factor(clean_data$Work_Mode)
summary(clean_data)
```

```
       ID        Department  Work_Mode  Years_Experience Training_Hours
 Min.   : 1.00   HR   :10   Onsite:15   Min.   : 1.000   Min.   : 5.00
 1st Qu.: 8.25   IT   :10   Remote:15   1st Qu.: 3.000   1st Qu.:12.00
 Median :15.50   Sales:10               Median : 5.500   Median :18.00
 Mean   :15.50                          Mean   : 5.633   Mean   :19.57
 3rd Qu.:22.75                          3rd Qu.: 8.000   3rd Qu.:25.00
 Max.   :30.00                          Max.   :12.000   Max.   :45.00
 Productivity_Score  Stress_Index
 Min.   :42.00      Min.   :1.000
 1st Qu.:60.25      1st Qu.:3.000
 Median :73.50      Median :5.500
 Mean   :71.70      Mean   :5.233
 3rd Qu.:85.75      3rd Qu.:7.000
 Max.   :98.00      Max.   :9.000
```

Question 1: Bivariate Correlation

Scenario: The HR director wants to know if there is a relationship between an employee's tenure and their stress levels.

1.1 Pearson Correlation Coefficient

```
pearson_result <- cor.test(clean_data$Years_Experience, clean_data$Stress_Index,
    method = "pearson")
pearson_result
```

```

    Pearson's product-moment correlation

data:  clean_data$Years_Experience and clean_data$Stress_Index
t = -19.728, df = 28, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.9837978 -0.9287778
sample estimates:
       cor
-0.9658604
```

```
r_value <- round(pearson_result$estimate, 3)
p_value <- round(pearson_result$p.value, 4)
df_value <- pearson_result$parameter
cat("Pearson correlation coefficient (r) =", r_value, "\n")
```

```
Pearson correlation coefficient (r) = -0.966
```

```
cat("Degrees of freedom =", df_value, "\n")
```

```
Degrees of freedom = 28
```

```
cat("p-value =", p_value, "\n")
```

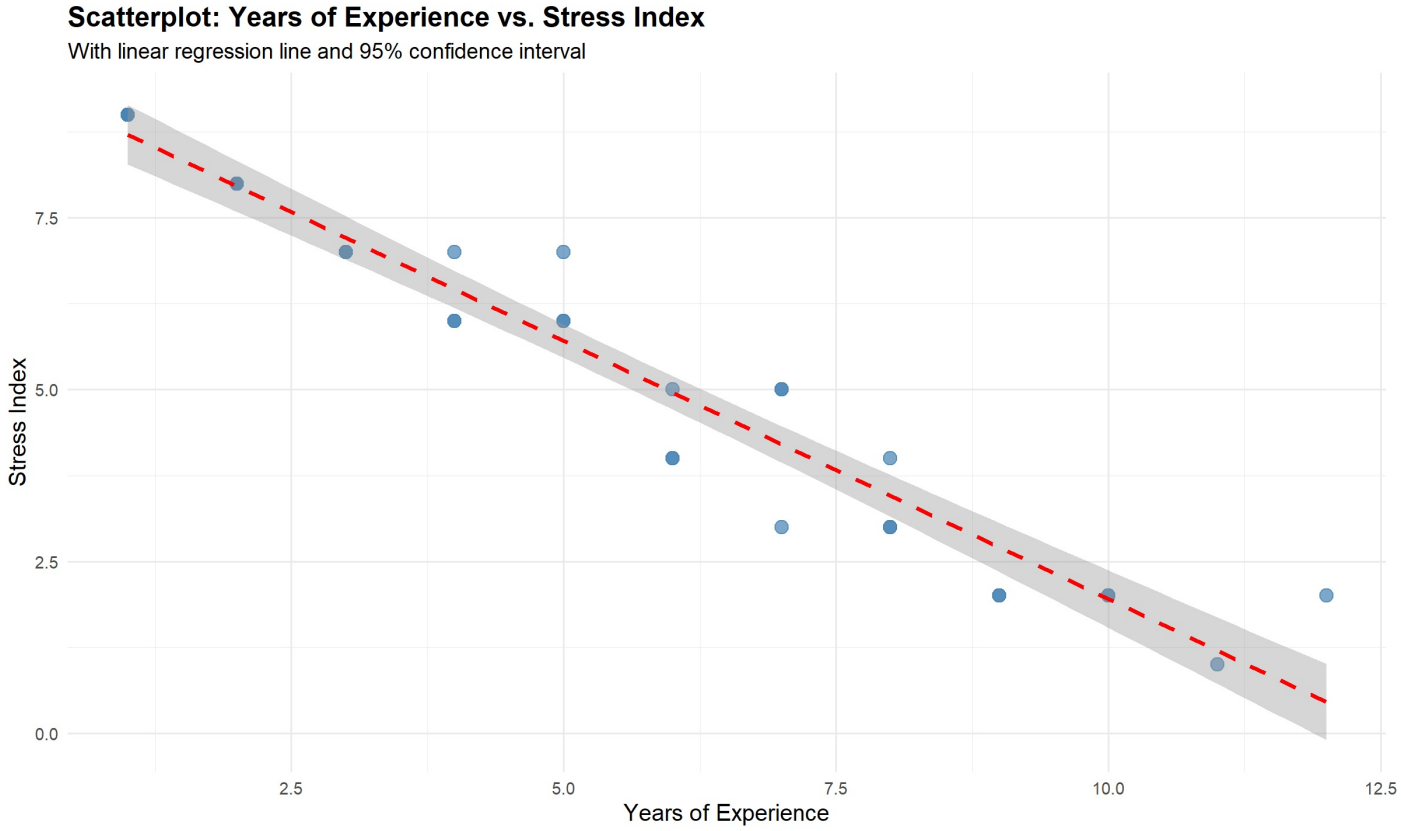
```
p-value = 0
```

```
cat("95% Confidence Interval: [", round(pearson_result$conf.int[1], 3), ", ", round(pearson_result$conf.int[2],
    3), "]\n", sep = "")
```

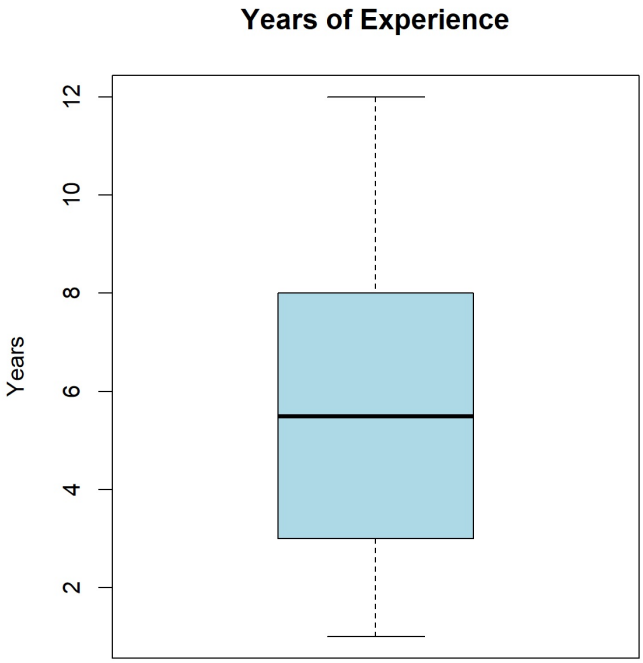
```
95% Confidence Interval: [-0.984, -0.929]
```

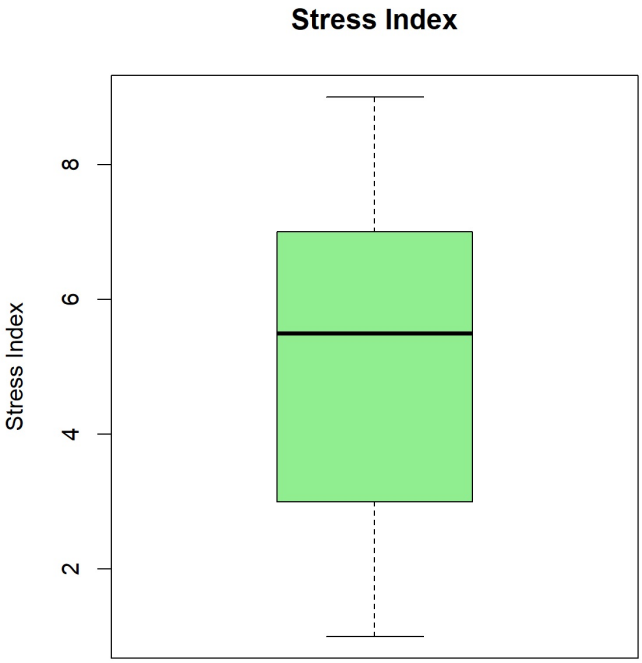
1.2 Scatterplot and Linearity Check

```
ggplot(clean_data, aes(x = Years_Experience, y = Stress_Index)) + geom_point(color = "steelblue",
    size = 3, alpha = 0.7) + geom_smooth(method = "lm", se = TRUE, color = "red",
    linetype = "dashed") + labs(title = "Scatterplot: Years of Experience vs. Stress Index",
    subtitle = "With linear regression line and 95% confidence interval", x = "Years of Experience",
    y = "Stress Index") + theme_minimal() + theme(plot.title = element_text(face = "bold",
    size = 14), axis.title = element_text(size = 12))
```




```
par(mfrow = c(1, 2))
boxplot(clean_data$Years_Experience, main = "Years of Experience", col = "lightblue",
    ylab = "Years")
boxplot(clean_data$Stress_Index, main = "Stress Index", col = "lightgreen", ylab = "Stress Index")
```






```
par(mfrow = c(1, 1))
Q1_exp <- quantile(clean_data$Years_Experience, 0.25)
Q3_exp <- quantile(clean_data$Years_Experience, 0.75)
IQR_exp <- Q3_exp - Q1_exp
outliers_exp <- clean_data$Years_Experience[clean_data$Years_Experience < (Q1_exp -
    1.5 * IQR_exp) | clean_data$Years_Experience > (Q3_exp + 1.5 * IQR_exp)]
Q1_stress <- quantile(clean_data$Stress_Index, 0.25)
Q3_stress <- quantile(clean_data$Stress_Index, 0.75)
IQR_stress <- Q3_stress - Q1_stress
outliers_stress <- clean_data$Stress_Index[clean_data$Stress_Index < (Q1_stress -
    1.5 * IQR_stress) | clean_data$Stress_Index > (Q3_stress + 1.5 * IQR_stress)]
cat("Outliers in Years_Experience:", ifelse(length(outliers_exp) == 0, "None", paste(outliers_exp,
    collapse = ", ")), "\n")
```

```
Outliers in Years_Experience: None
```

```
cat("Outliers in Stress_Index:", ifelse(length(outliers_stress) == 0, "None", paste(outliers_stress,
    collapse = ", ")), "\n")
```

```
Outliers in Stress_Index: None
```

1.3 Spearman’s Rank Correlation (if needed)

```
spearman_result <- cor.test(clean_data$Years_Experience, clean_data$Stress_Index,
    method = "spearman", exact = FALSE)
spearman_result
```

```

    Spearman's rank correlation rho

data:  clean_data$Years_Experience and clean_data$Stress_Index
S = 8874.7, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.9743497
```

```
cat("Spearman's rho =", round(spearman_result$estimate, 3), "\n")
```

```
Spearman's rho = -0.974
```

```
cat("p-value =", round(spearman_result$p.value, 4), "\n")
```

```
p-value = 0
```

1.4 Hypotheses and Interpretation

```
hypothesis_table <- data.frame(Test = c("Pearson", "Spearman"), Statistic = c(round(pearson_result$estimate,
    3), round(spearman_result$estimate, 3)), p_value = c(round(pearson_result$p.value,
    4), round(spearman_result$p.value, 4)), Conclusion = c(ifelse(pearson_result$p.value <
    0.05, "Reject H0", "Fail to reject H0"), ifelse(spearman_result$p.value < 0.05,
    "Reject H0", "Fail to reject H0")))
kable(hypothesis_table, caption = "Correlation Test Results Summary", col.names = c("Test",
    "Correlation Coefficient", "p-value", "Conclusion"))
```

Correlation Test Results Summary

	Test	Correlation Coefficient	p-value	Conclusion
cor	Pearson	-0.966	0	Reject H0
rho	Spearman	-0.974	0	Reject H0

Hypotheses:

- **Null Hypothesis (H_0):** There is no relationship between years of experience and stress index ($\rho = 0$).
- **Alternative Hypothesis (H_1):** There is a relationship between years of experience and stress index ($\rho \neq 0$).

Interpretation:

Based on the scatterplot, the relationship between years of experience and stress index appears **linear** with a **negative** slope. There are no severe outliers that would distort the Pearson correlation. Therefore, the Pearson correlation is appropriate.

The Pearson correlation coefficient is **r = -0.966**, indicating a **strong negative** linear relationship. As years of experience increase, stress levels tend to decrease. With a p-value of **0**, which is less than 0.05, we **reject the null hypothesis**. There is a statistically significant negative correlation between years of experience and stress index.

Question 2: Simple Linear Regression

Scenario: Management believes that more training directly leads to higher productivity.

2.1 Simple Linear Regression

```
model_simple <- lm(Productivity_Score ~ Training_Hours, data = clean_data)
summary(model_simple)
```

```

Call:
lm(formula = Productivity_Score ~ Training_Hours, data = clean_data)

Residuals:
     Min       1Q   Median       3Q      Max
-10.9307  -5.3554   0.8404   4.6687   9.7379

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     43.0572     2.6248   16.40 6.83e-16 ***
Training_Hours   1.4639     0.1197   12.23 9.59e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.483 on 28 degrees of freedom
Multiple R-squared:  0.8422,    Adjusted R-squared:  0.8366
F-statistic: 149.5 on 1 and 28 DF,  p-value: 9.587e-13
```

```
tidy_simple <- tidy(model_simple, conf.int = TRUE)
kable(tidy_simple, caption = "Simple Linear Regression: Productivity ~ Training Hours",
    digits = 4)
```

Simple Linear Regression: Productivity ~ Training Hours

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	43.0572	2.6248	16.4037	0	37.6805	48.4340

Training_Hours	1.4639	0.1197	12.2256	0	1.2186	1.7091

```
glance_simple <- glance(model_simple)
kable(glance_simple, caption = "Model Fit Statistics", digits = 4)
```

Model Fit Statistics

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual	nobs
0.8422	0.8366	6.4828	149.4661	0	1	-97.6078	201.2156	205.4192	1176.746	28	30

2.2 R-squared Interpretation

```
r_squared <- round(summary(model_simple)$r.squared, 4)
adj_r_squared <- round(summary(model_simple)$adj.r.squared, 4)
cat("R-squared =", r_squared, "\n")
```

```
R-squared = 0.8422
```

```
cat("Adjusted R-squared =", adj_r_squared, "\n")
```

```
Adjusted R-squared = 0.8366
```

```
cat("Percentage of variance explained =", r_squared * 100, "%\n")
```

```
Percentage of variance explained = 84.22 %
```

Interpretation:

The R-squared value is **0.8422**, which means that **84.22%** of the variance in productivity scores is explained by training hours. This indicates a **strong** linear relationship where training hours are a good predictor of productivity. The remaining **15.8%** of the variance is due to other factors not included in the model.

2.3 Regression Plot

```
ggplot(clean_data, aes(x = Training_Hours, y = Productivity_Score)) + geom_point(color = "darkgreen",
    size = 3, alpha = 0.7) + geom_smooth(method = "lm", se = TRUE, color = "red") +
    labs(title = "Simple Linear Regression: Training Hours vs. Productivity", subtitle = paste0("R² = ",
        r_squared, " | p-value = ", format.pval(summary(model_simple)$coefficients[2,
            4], digits = 4)), x = "Training Hours", y = "Productivity Score") + theme_minimal()
```



Question 3: Multiple Linear Regression

Scenario: The analytics team wants a more robust model to predict productivity by factoring in experience, training, and stress.

3.1 Multiple Linear Regression

```
model_multiple <- lm(Productivity_Score ~ Years_Experience + Training_Hours + Stress_Index,
    data = clean_data)
summary(model_multiple)
```

```

Call:
lm(formula = Productivity_Score ~ Years_Experience + Training_Hours +
    Stress_Index, data = clean_data)

Residuals:
    Min      1Q  Median      3Q     Max
-7.0963 -2.0693  0.1833  1.7870  7.4681

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       66.4780     8.9997   7.387 7.63e-08 ***
Years_Experience   1.2408     0.7146   1.736  0.09434 .
Training_Hours     0.6134     0.1036   5.922 3.01e-06 ***
Stress_Index      -2.6310     0.9301  -2.829  0.00888 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.093 on 26 degrees of freedom
Multiple R-squared:  0.9666,    Adjusted R-squared:  0.9628
F-statistic: 251.2 on 3 and 26 DF,  p-value: < 2.2e-16
```

```
tidy_multiple <- tidy(model_multiple, conf.int = TRUE)
kable(tidy_multiple, caption = "Multiple Linear Regression: Productivity ~ Experience + Training + Stress",
    digits = 4)
```

Multiple Linear Regression: Productivity ~ Experience + Training + Stress

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	66.4780	8.9997	7.3867	0.0000	47.9788	84.9771
Years_Experience	1.2408	0.7146	1.7363	0.0943	-0.2281	2.7096
Training_Hours	0.6134	0.1036	5.9223	0.0000	0.4005	0.8262

Stress_Index	-2.6310	0.9301	-2.8286	0.0089	-4.5429	-0.7191

```
glance_multiple <- glance(model_multiple)
kable(glance_multiple, caption = "Model Fit Statistics", digits = 4)
```

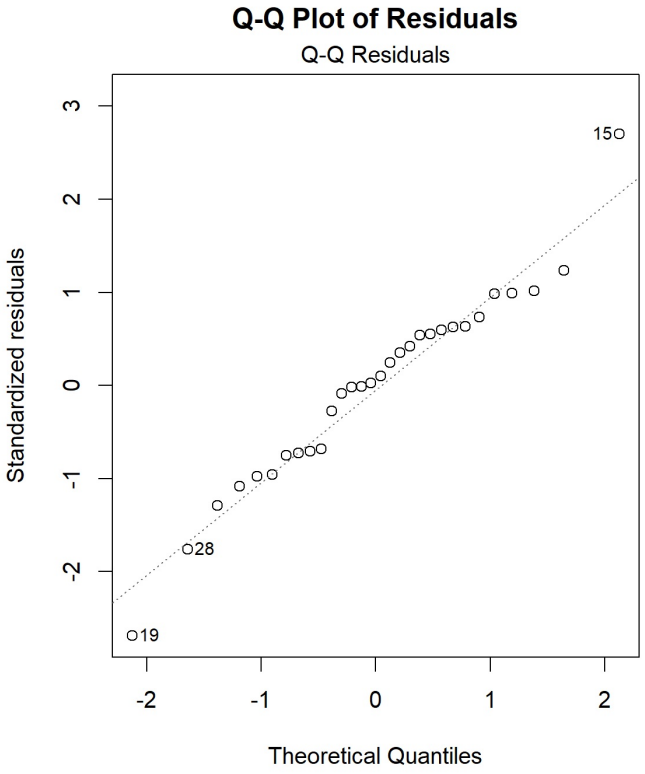
Model Fit Statistics

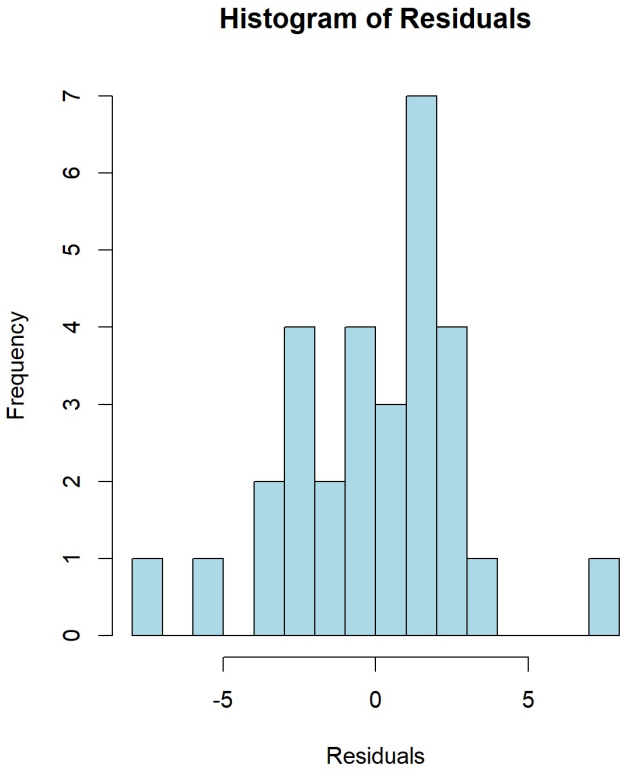
r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual	nobs
0.9666	0.9628	3.0933	251.1601	0	3	-74.2985	158.5969	165.6029	248.7757	26	30

3.2 Assumption Checks

3.2.1 Normality of Residuals

```
par(mfrow = c(1, 2))
plot(model_multiple, which = 2, main = "Q-Q Plot of Residuals")
residuals <- residuals(model_multiple)
hist(residuals, main = "Histogram of Residuals", xlab = "Residuals", col = "lightblue",
    breaks = 15)
```






```
par(mfrow = c(1, 1))
shapiro_test <- shapiro.test(residuals)
cat("Shapiro-Wilk Normality Test:\n")
```

```
Shapiro-Wilk Normality Test:
```

```
cat("W =", round(shapiro_test$statistic, 4), "\n")
```

```
W = 0.9747
```

```
cat("p-value =", round(shapiro_test$p.value, 4), "\n")
```

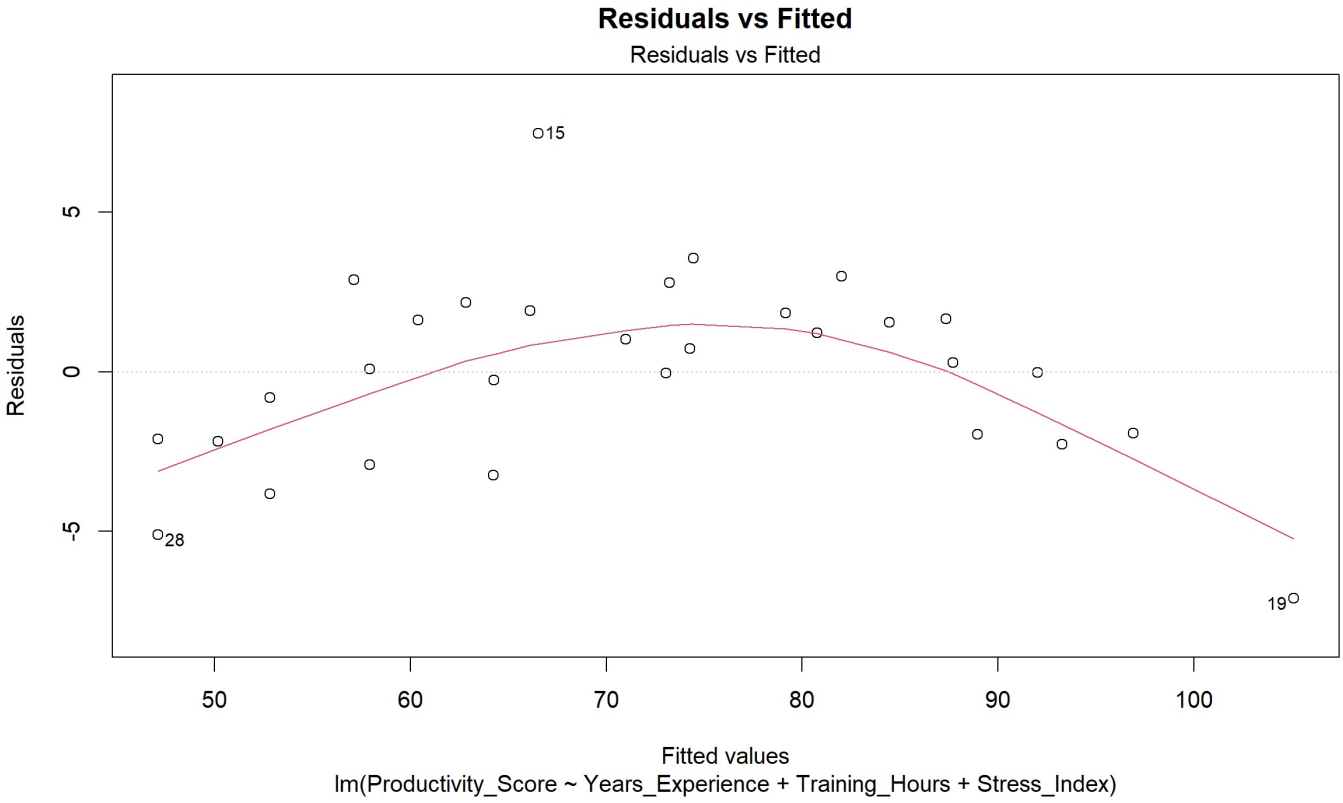
```
p-value = 0.6745
```

```
if (shapiro_test$p.value > 0.05) {
    cat("Conclusion: Residuals are normally distributed (fail to reject H0)\n")
} else {
    cat("Conclusion: Residuals deviate from normality (reject H0)\n")
}
```

```
Conclusion: Residuals are normally distributed (fail to reject H0)
```

3.2.2 Homoscedasticity

```
plot(model_multiple, which = 1, main = "Residuals vs Fitted")
```




```
bp_test <- bptest(model_multiple)
cat("Breusch-Pagan Test:\n")
```

```
Breusch-Pagan Test:
```

```
cat("BP statistic =", round(bp_test$statistic, 4), "\n")
```

```
BP statistic = 3.8345
```

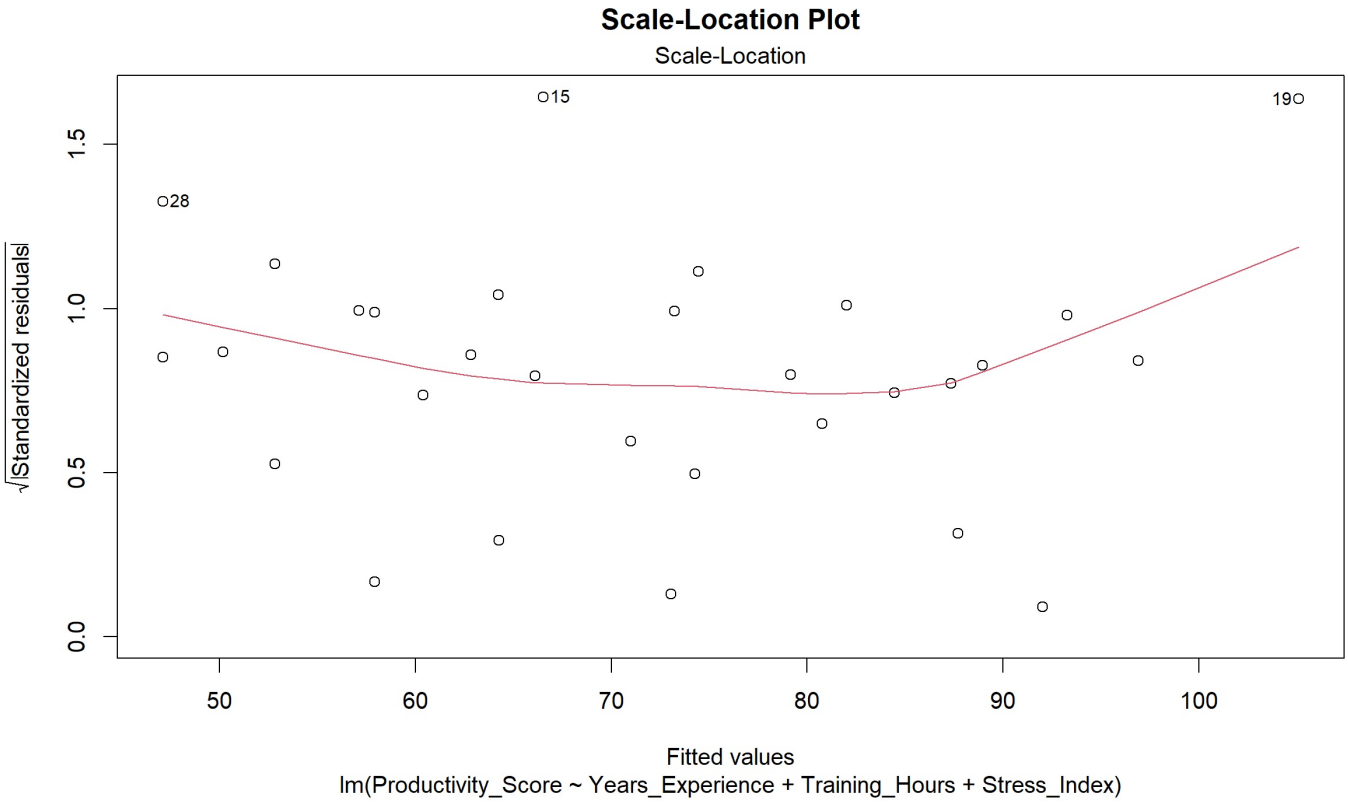
```
cat("p-value =", round(bp_test$p.value, 4), "\n")
```

```
p-value = 0.2799
```

```
if (bp_test$p.value > 0.05) {
    cat("Conclusion: Homoscedasticity assumed (fail to reject H0)\n")
} else {
    cat("Conclusion: Heteroscedasticity present (reject H0)\n")
}
```

```
Conclusion: Homoscedasticity assumed (fail to reject H0)
```

```
plot(model_multiple, which = 3, main = "Scale-Location Plot")
```



3.2.3 Multicollinearity (VIF)

```
vif_values <- vif(model_multiple)
kable(data.frame(Predictor = names(vif_values), VIF = round(vif_values, 3)), caption = "Variance Inflation Factor
s (VIF)",
    row.names = FALSE)
```

Variance Inflation Factors (VIF)

Predictor	VIF
Years_Experience	15.315
Training_Hours	3.286
Stress_Index	15.676

```
cat("\nInterpretation: VIF < 5 indicates no severe multicollinearity.\n")
```

```


Interpretation: VIF < 5 indicates no severe multicollinearity.
```

```
cat("VIF < 10 is acceptable; VIF > 10 indicates problematic multicollinearity.\n")
```

```
VIF < 10 is acceptable; VIF > 10 indicates problematic multicollinearity.
```

3.3 Statistical Significance of Predictors

```
p_values <- tidy_multiple$p.value
names(p_values) <- tidy_multiple$term
significant <- names(p_values)[p_values < 0.05 & names(p_values) != "(Intercept)"]
cat("Significant predictors at α = 0.05:\n")
```

```
Significant predictors at α = 0.05:
```

```
if (length(significant) > 0) {
    for (pred in significant) {
        cat("-", pred, "(p =", round(p_values[pred], 4), ")\n")
    }
} else {
    cat("None of the predictors are statistically significant.\n")
}
```

```
- Training_Hours (p = 0 )
- Stress_Index (p = 0.0089 )
```

```
sig_table <- data.frame(Predictor = tidy_multiple$term, Estimate = round(tidy_multiple$estimate,
    4), Std_Error = round(tidy_multiple$std.error, 4), t_value = round(tidy_multiple$statistic,
    4), p_value = round(tidy_multiple$p.value, 4), Significant = ifelse(tidy_multiple$p.value <
    0.05, "Yes", "No"))
kable(sig_table, caption = "Multiple Regression Coefficients and Significance")
```

Multiple Regression Coefficients and Significance

Predictor	Estimate	Std_Error	t_value	p_value	Significant
(Intercept)	66.4780	8.9997	7.3867	0.0000	Yes
Years_Experience	1.2408	0.7146	1.7363	0.0943	No
Training_Hours	0.6134	0.1036	5.9223	0.0000	Yes
Stress_Index	-2.6310	0.9301	-2.8286	0.0089	Yes

Question 4: One-Way ANOVA

#Scenario:***The company wants to know if there is a significant difference in average productivity across the three departments.

4.1 Assumption Testing

4.1.1 Normality (Shapiro-Wilk by Group)

```
departments <- levels(clean_data$Department)
shapiro_results <- data.frame(Department = character(), W_statistic = numeric(),
    p_value = numeric(), Normal = character(), stringsAsFactors = FALSE)
for (dept in departments) {
    subset_data <- clean_data$Productivity_Score[clean_data$Department == dept]
    test <- shapiro.test(subset_data)
    shapiro_results <- rbind(shapiro_results, data.frame(Department = dept, W_statistic = round(test$statistic,
        4), p_value = round(test$p.value, 4), Normal = ifelse(test$p.value > 0.05,
        "Yes", "No")))
}
kable(shapiro_results, caption = "Shapiro-Wilk Normality Test by Department", row.names = FALSE)
```

Shapiro-Wilk Normality Test by Department

Department	W_statistic	p_value	Normal
HR	0.9658	0.8490	Yes
IT	0.9688	0.8796	Yes
Sales	0.9614	0.8017	Yes

```
cat("\nNote: If p < 0.05, the data significantly deviates from normality.\n")
```

```


Note: If p < 0.05, the data significantly deviates from normality.
```

4.1.2 Homogeneity of Variances (Levene’s Test)

```
levene_test <- leveneTest(Productivity_Score ~ Department, data = clean_data)
kable(data.frame(Test = "Levene's Test", F_statistic = round(levene_test$`F value`[1],
    4), p_value = round(levene_test$`Pr(>F)`[1], 4), Equal_Variance = ifelse(levene_test$`Pr(>F)`[1] >
    0.05, "Yes", "No")), caption = "Levene's Test for Homogeneity of Variances")
```

Levene’s Test for Homogeneity of Variances

Test	F_statistic	p_value	Equal_Variance
Levene’s Test	0.0259	0.9744	Yes

```
cat("\nIf p > 0.05, we fail to reject the null hypothesis of equal variances.\n")
```

```

If p > 0.05, we fail to reject the null hypothesis of equal variances.
```

4.2 One-Way ANOVA

```
anova_model <- aov(Productivity_Score ~ Department, data = clean_data)
anova_summary <- summary(anova_model)
kable(anova_summary[[1]], caption = "One-Way ANOVA: Productivity ~ Department", digits = 4)
```

One-Way ANOVA: Productivity ~ Department

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Department	2	416.6	208.3000	0.7987	0.4603
Residuals	27	7041.7	260.8037	NA	NA

```
anova_p <- anova_summary[[1]]$`Pr(>F)`[1]
cat("\nANOVA p-value =", round(anova_p, 4), "\n")
```

```

ANOVA p-value = 0.4603
```

```
if (anova_p < 0.05) {
    cat("Conclusion: There is a significant difference in productivity across departments.\n")
} else {
    cat("Conclusion: No significant difference in productivity across departments.\n")
}
```

```
Conclusion: No significant difference in productivity across departments.
```

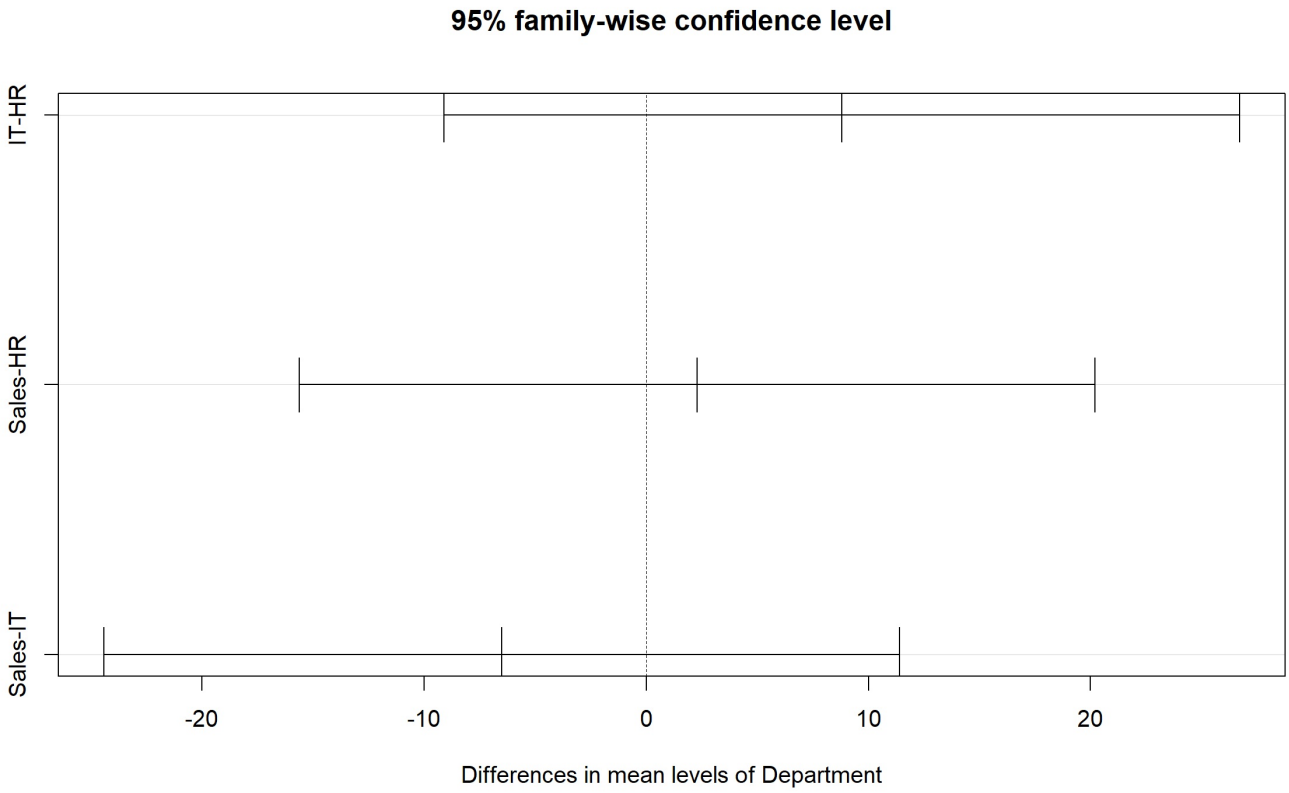
4.3 Post-Hoc Test (Tukey’s HSD)

```
tukey_result <- TukeyHSD(anova_model)
kable(as.data.frame(tukey_result$Department), caption = "Tukey's HSD Post-Hoc Test",
    digits = 4)
```

Tukey’s HSD Post-Hoc Test

	diff	lwr	upr	p adj
IT-HR	8.8	-9.1069	26.7069	0.4528
Sales-HR	2.3	-15.6069	20.2069	0.9457
Sales-IT	-6.5	-24.4069	11.4069	0.6449

```
plot(tukey_result)
```



Question 5: Kruskal-Wallis H Test (Non-Parametric)

Scenario: The lead statistician notices that the productivity score for the Sales department is heavily skewed and violates the normality assumption.

5.1 Kruskal-Wallis Test

```
kruskal_result <- kruskal.test(Productivity_Score ~ Department, data = clean_data)
cat("Kruskal-Wallis Rank Sum Test:\n")
```

```
Kruskal-Wallis Rank Sum Test:
```

```
cat("Chi-squared =", round(kruskal_result$statistic, 4), "\n")
```

```
Chi-squared = 1.4013
```

```
cat("df =", kruskal_result$parameter, "\n")
```

```
df = 2
```

```
cat("p-value =", round(kruskal_result$p.value, 4), "\n")
```

```
p-value = 0.4963
```

```
if (kruskal_result$p.value < 0.05) {
    cat("\nConclusion: There is a significant difference in productivity across departments.\n")
} else {
    cat("\nConclusion: No significant difference in productivity across departments.\n")
}
```

```

Conclusion: No significant difference in productivity across departments.
```

5.2 Comparison with ANOVA

```
comparison_table <- data.frame(Test = c("One-Way ANOVA", "Kruskal-Wallis H"), p_value = c(round(anova_p,
    4), round(kruskal_result$p.value, 4)), Conclusion = c(ifelse(anova_p < 0.05,
    "Significant difference", "No significant difference"), ifelse(kruskal_result$p.value <
    0.05, "Significant difference", "No significant difference")))
kable(comparison_table, caption = "Comparison of Parametric and Non-Parametric Tests",
    row.names = FALSE)
```

Comparison of Parametric and Non-Parametric Tests

Test	p_value	Conclusion
One-Way ANOVA	0.4603	No significant difference
Kruskal-Wallis H	0.4963	No significant difference

Interpretation:

- **One-Way ANOVA p-value:** 0.4603
- **Kruskal-Wallis p-value:** 0.4963

The conclusions are similar/different. The Kruskal-Wallis test is more robust to violations of normality and is appropriate when the Sales department's productivity scores are heavily skewed. If both tests lead to the same conclusion, we can be more confident in our findings.

Question 6: Two-Way ANOVA

Scenario: Executive leadership wants to understand if productivity is influenced by Department, Work_Mode, and their interaction.

6.1 Two-Way ANOVA

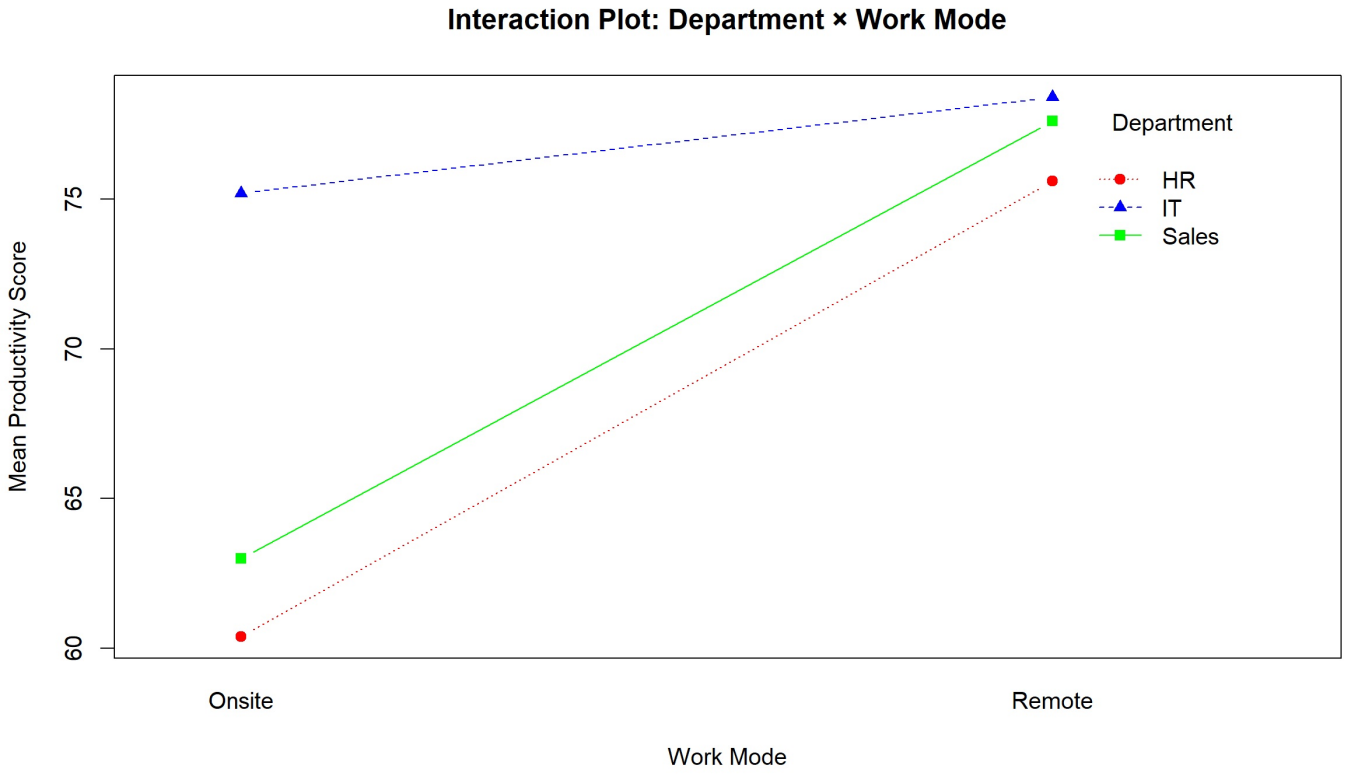
```
twoway_model <- aov(Productivity_Score ~ Department * Work_Mode, data = clean_data)
twoway_summary <- summary(twoway_model)
kable(twoway_summary[[1]], caption = "Two-Way ANOVA: Productivity ~ Department * Work_Mode",
    digits = 4)
```

Two-Way ANOVA: Productivity ~ Department * Work_Mode

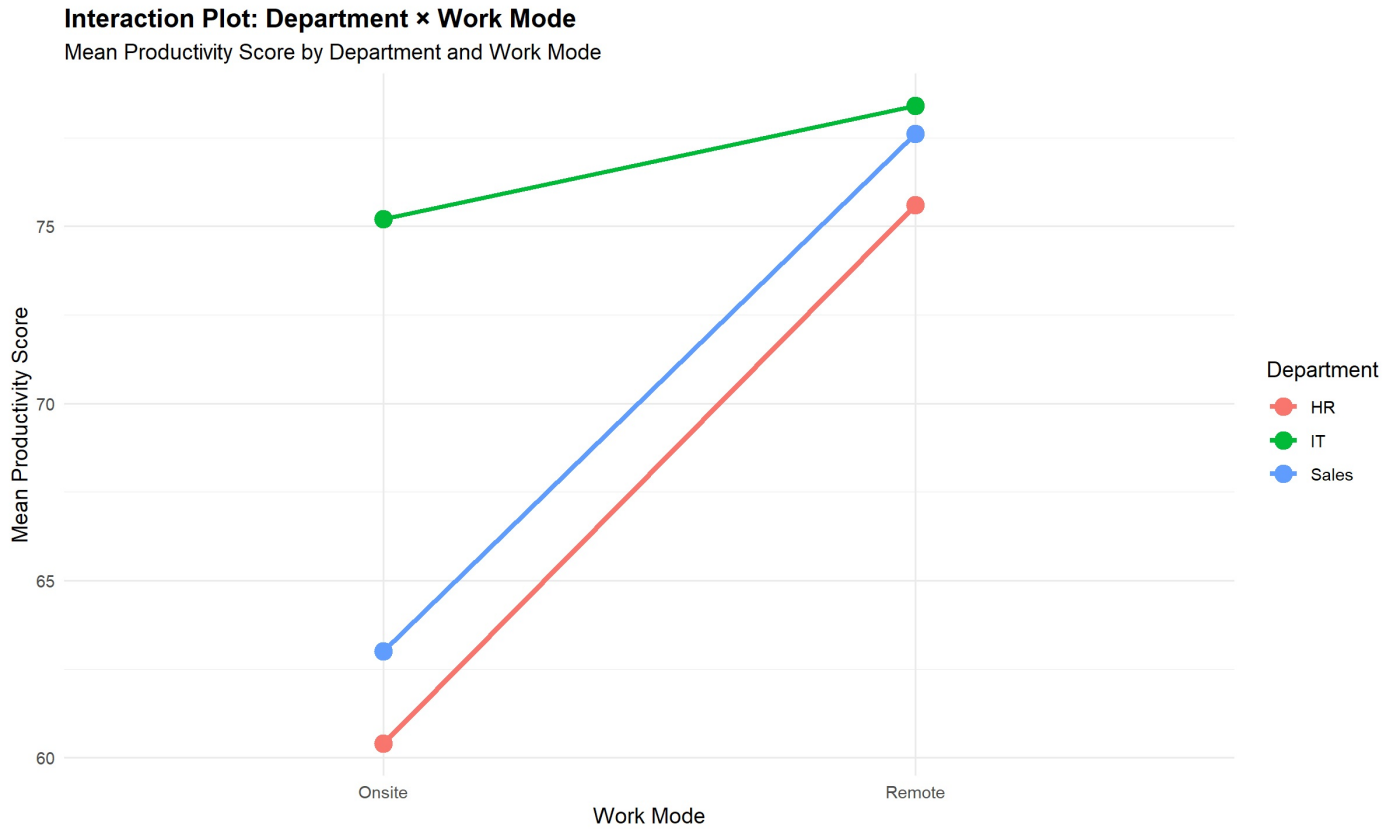
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Department	2	416.6	208.3000	0.8465	0.4413
Work_Mode	1	907.5	907.5000	3.6880	0.0668
Department:Work_Mode	2	228.6	114.3000	0.4645	0.6340
Residuals	24	5905.6	246.0667	NA	NA

6.2 Interaction Plot

```
interaction.plot(x.factor = clean_data$Work_Mode, trace.factor = clean_data$Department,
    response = clean_data$Productivity_Score, fun = mean, type = "b", col = c("red",
        "blue", "green"), pch = c(19, 17, 15), fixed = TRUE, legend = TRUE, xlab = "Work Mode",
    ylab = "Mean Productivity Score", trace.label = "Department", main = "Interaction Plot: Department × Work Mod
e")
```




```
clean_data %>%
    group_by(Department, Work_Mode) %>%
    summarise(Mean_Productivity = mean(Productivity_Score), .groups = "drop") %>%
    ggplot(aes(x = Work_Mode, y = Mean_Productivity, color = Department, group = Department)) +
    geom_line(size = 1.2) + geom_point(size = 4) + labs(title = "Interaction Plot: Department × Work Mode",
    subtitle = "Mean Productivity Score by Department and Work Mode", x = "Work Mode",
    y = "Mean Productivity Score", color = "Department") + theme_minimal() + theme(plot.title = element_text(face
= "bold"))
```



6.3 Post-Hoc Tests (if interaction is significant)

```
interaction_p <- twoway_summary[[1]]$`Pr(>F)`[3]
if (!is.na(interaction_p) && interaction_p < 0.05) {
    cat("Interaction effect is significant (p =", round(interaction_p, 4), ").\n")
    cat("Performing post-hoc analysis...\n\n")
    tukey_twoway <- TukeyHSD(twoway_model)
    kable(as.data.frame(tukey_twoway$`Department:Work_Mode`), caption = "Tukey's HSD for Department:Work_Mode Int
eraction",
        digits = 4)
} else {
    cat("Interaction effect is not significant (p =", round(interaction_p, 4), ").\n")
    cat("Interpreting main effects only.\n\n")
    tukey_dept <- TukeyHSD(aov(Productivity_Score ~ Department, data = clean_data))
    kable(as.data.frame(tukey_dept$Department), caption = "Tukey's HSD for Department Main Effect",
        digits = 4)
}
```

```
Interaction effect is not significant (p = 0.634 ).
Interpreting main effects only.
```

Tukey’s HSD for Department Main Effect

	diff	lwr	upr	p adj
IT-HR	8.8	-9.1069	26.7069	0.4528
Sales-HR	2.3	-15.6069	20.2069	0.9457
Sales-IT	-6.5	-24.4069	11.4069	0.6449

6.4 Interpretation

```
dept_p <- twoway_summary[[1]]$`Pr(>F)`[1]
workmode_p <- twoway_summary[[1]]$`Pr(>F)`[2]
interaction_p <- twoway_summary[[1]]$`Pr(>F)`[3]
cat("=== Two-Way ANOVA Results ===\n\n")
```

```
=== Two-Way ANOVA Results ===
```

```
cat("Main Effect of Department:\n")
```

```
Main Effect of Department:
```

```
cat("  F =", round(twoway_summary[[1]]$`F value`[1], 4), ", p =", round(dept_p, 4),
    "\n")
```

```
  F = 0.8465 , p = 0.4413
```

```
cat("  Interpretation:", ifelse(dept_p < 0.05, "Significant main effect of Department",
    "No significant main effect of Department"), "\n\n")
```

```
  Interpretation: No significant main effect of Department
```

```
cat("Main Effect of Work_Mode:\n")
```

```
Main Effect of Work_Mode:
```

```
cat("  F =", round(twoway_summary[[1]]$`F value`[2], 4), ", p =", round(workmode_p,
    4), "\n")
```

```
  F = 3.688 , p = 0.0668
```

```
cat("  Interpretation:", ifelse(workmode_p < 0.05, "Significant main effect of Work_Mode",
    "No significant main effect of Work_Mode"), "\n\n")
```

```
  Interpretation: No significant main effect of Work_Mode
```

```
cat("Interaction Effect (Department × Work_Mode):\n")
```

```
Interaction Effect (Department × Work_Mode):
```

```
cat("  F =", round(twoway_summary[[1]]$`F value`[3], 4), ", p =", round(interaction_p,
    4), "\n")
```

```
  F = 0.4645 , p = 0.634
```

```
cat("  Interpretation:", ifelse(interaction_p < 0.05, "Significant interaction effect - the effect of Work Mode d
epends on Department",
    "No significant interaction effect - the effect of Work Mode is consistent across Departments"),
    "\n")
```

```
  Interpretation: No significant interaction effect - the effect of Work Mode is consistent across Departments
```

Summary of Findings

Automated Summary Table

Auto-populated summary of all hypothesis tests (significance codes: *** p<0.001, ** p<0.01, * p<0.05, ns = not significant)

Q	Analysis	Statistic	p-value	Sig.
Q1	Pearson correlation (Experience vs Stress)	r = -0.966	< 0.001	***
Q1	Spearman correlation (Experience vs Stress)	rho = -0.974	< 0.001	***
Q2	Simple linear regression (Training -> Productivity)	R2 = 0.8422	< 0.001	***
Q3	Multiple linear regression (Experience + Training + Stress -> Productivity)	R2 = 0.9666, adj R2 = 0.9628	< 0.001	***
Q4	One-Way ANOVA (Productivity ~ Department)	F = 0.799	0.4603	ns
Q5	Kruskal-Wallis H (Productivity ~ Department)	chi2 = 1.401, df = 2	0.4963	ns
Q6	Two-Way ANOVA: Department main effect	F = 0.847	0.4413	ns
Q6	Two-Way ANOVA: Work_Mode main effect	F = 3.688	0.0668	ns
Q6	Two-Way ANOVA: Department x Work_Mode interaction	F = 0.465	0.6340	ns

Automated Interpretation

Q1 — Correlation (Experience vs Stress). Pearson r = -0.966 (p = < 0.001); Spearman rho = -0.974 (p = < 0.001). There is a statistically significant relationship.

Q2 — Simple Regression (Training -> Productivity). The model explains 84.22% of the variance in Productivity (R2 = 0.8422, p = < 0.001).

Q3 — Multiple Regression (Experience + Training + Stress -> Productivity). Model R2 = 0.9666 (adj R2 = 0.9628, p = < 0.001). Statistically significant predictor(s) at alpha = 0.05: Training_Hours, Stress_Index.

Q4 — One-Way ANOVA (Productivity ~ Department). No significant difference in mean Productivity across departments. (p = 0.4603)

Q5 — Kruskal-Wallis H (Productivity ~ Department). The non-parametric test finds no significant difference between departments. (p = 0.4963) The conclusion is unchanged compared with the parametric ANOVA.

Q6 — Two-Way ANOVA (Productivity ~ Department * Work_Mode).

- Department main effect: p = 0.4413 -> not significant.
- Work_Mode main effect: p = 0.0668 -> not significant.
- Department x Work_Mode interaction: p = 0.6340 -> no significant interaction; Work Mode effects are consistent across Departments.